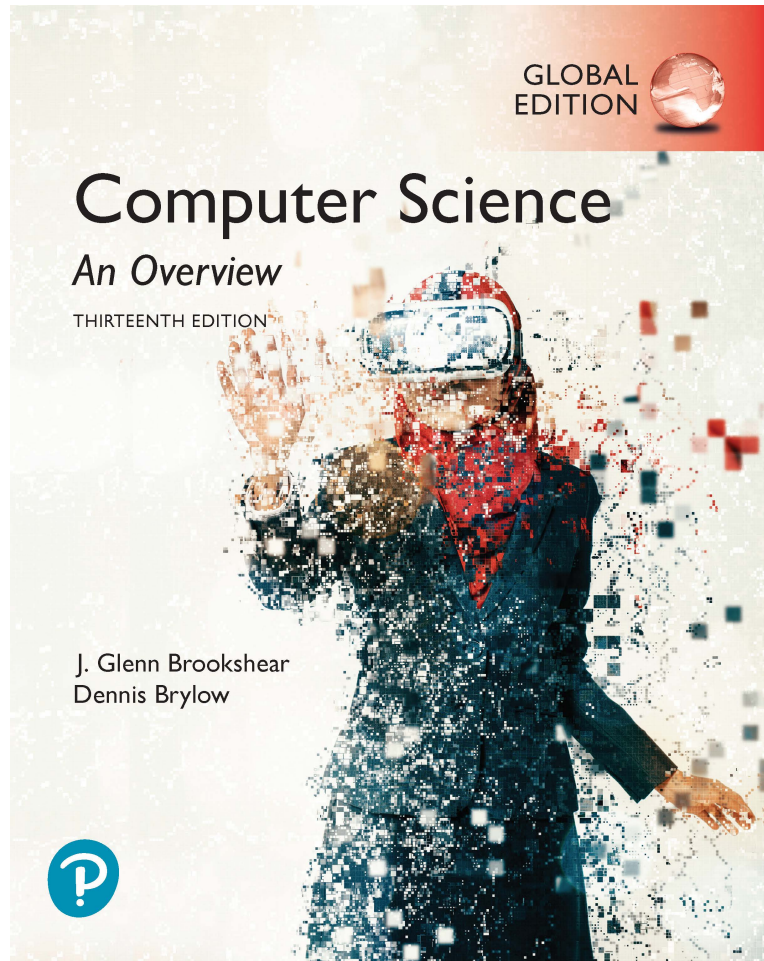


Computer Science An Overview

13th Edition, Global Edition



Chapter 9 Database Systems

資料庫系統

Chapter 9: Database Systems

- 9.1 Database Fundamentals
- 9.2 The Relational Model
- 9.3 Object-Oriented Databases
- 9.4 Maintaining Database Integrity
- 9.5 Traditional File Structures — 期末範圍
- 9.6 Data Mining
- 9.7 Social Impact of Database Technology

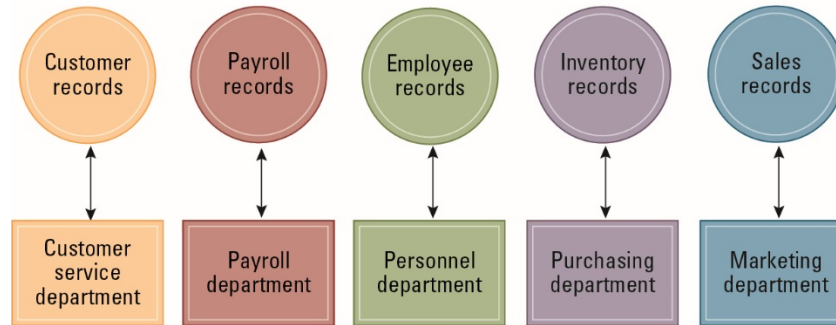
9.1 Database Fundamentals

Database: A collection of data that is multidimensional in the sense that internal links between its entries make the information accessible from a variety of perspectives

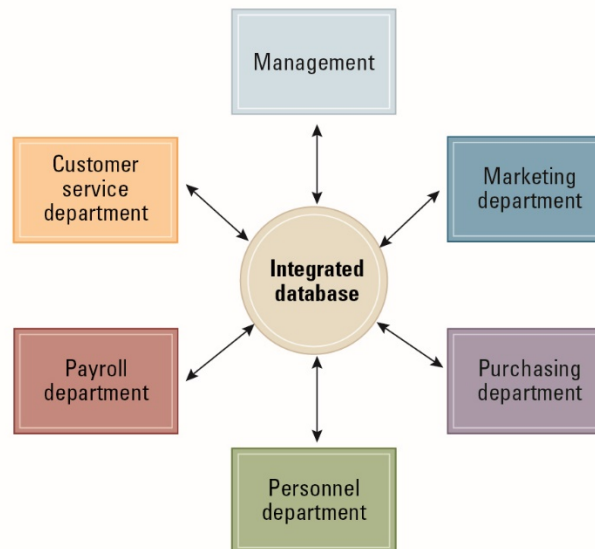
資料庫：一種 **多維度** 資料的集合，其內部條目之間的連結 s.t.
資訊能夠從多角度進行存取

Figure 9.1 A file versus a database organization

a. File-oriented information system



b. Database-oriented information system



The Role of Schemas

描述資料庫結構的工具，資料庫軟件用他來維護資料庫

- **Schema:** A description of the structure of an entire database, used by database software to maintain the database

- **Subschema:** A description of only that portion of the database pertinent to a particular user's needs, used to prevent sensitive data from being accessed by unauthorized personnel

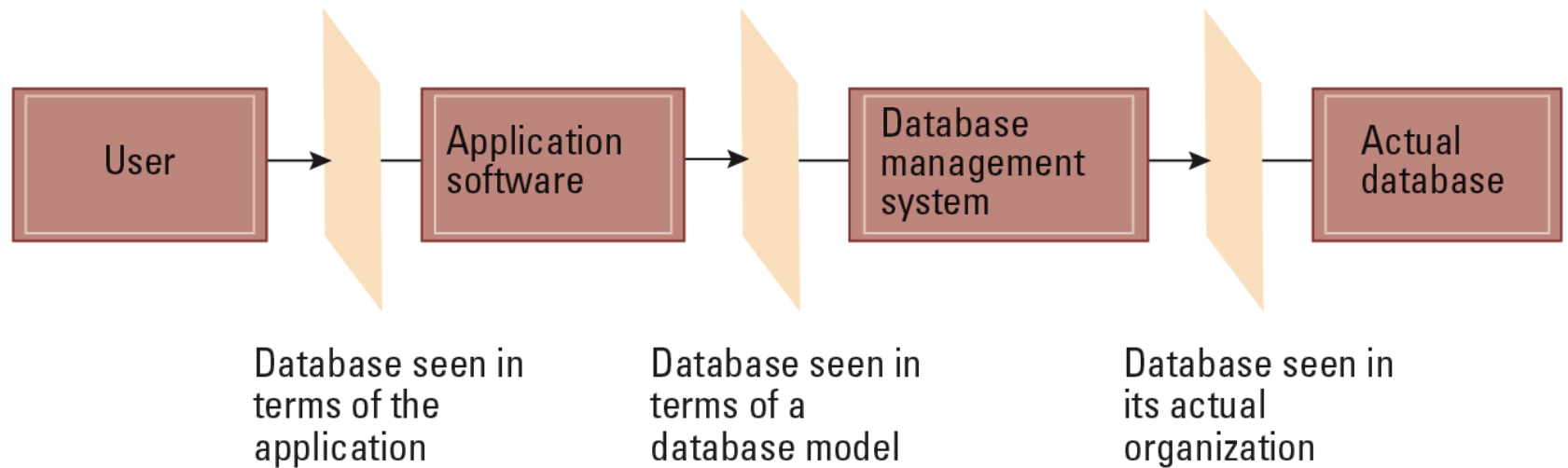
僅對使用者需要的資料庫進行描述，防止未經授權的人員存取敏感資料。

Database Management Systems

資料庫管理系統

- **Database Management System (DBMS):** A software layer that manipulates a database in response to requests from applications
負責回應程式對資料庫的請求
- **Distributed Database:** A database stored on multiple machines
分散式資料庫
被分散存儲在多台機器上的資料庫
- DBMS will mask this organizational detail from its users
資料獨立性
DBMS 會負責對 user 隱藏資料或關聯的訊息
- **Data independence:** The ability to change the organization of a database without changing the application software that uses it
*不是一件容易達成的事。
不需更改使用資料庫的應用程式即可更改資料庫的組織能力。

Figure 9.2 The conceptual layers of a database implementation



Database Models

- **Database model:** A conceptual view of a database
 - Relational database model 關聯式資料庫模型
 - Object-oriented database model 物件導向資料庫模型

9.2 The Relational Model

關聯

- **Relation:** A rectangular table
- 屬性 **Attribute:** A column in the table
- 數組 **Tuple:** A row in the table

Figure 9.3 A relation containing employee information

Empl Id	Name	Address	SSN
25X15	Joe E. Baker	33 Nowhere St.	111223333
34Y70	Cheryl H. Clark	563 Downtown Ave.	999009999
23Y34	G. Jerry Smith	1555 Circle Dr.	111005555
•	•	•	•
•	•	•	•
•	•	•	•

Issues of Relational Design

避免重複的 Concepts 在同個 relation 中出現

- Avoid multiple concepts within one relation
 - Can lead to redundant data
 - Deleting a tuple could also delete necessary but unrelated information

Improving a Relational Design

分解 (壓縮)

- **Decomposition:** Dividing the columns of a relation into two or more relations, duplicating those columns necessary to maintain relationships

無損壓縮

- **Lossless or nonloss decomposition:** A “correct” decomposition that does not lose any information

將關聯式欄位分成兩個 or 多個關聯式欄位，
將必要的欄位複製、使其易於維護

Figure 9.4 A relation containing redundancy

Empl Id	Name	Address	SSN	Job Id	Job Title	Skill Code	Dept	Start Date	Term Date
25X15	Joe E. Baker	33 Nowhere St.	111223333	F5	Floor manager	FM3	Sales	9-1-2009	9-30-2010
25X15	Joe E. Baker	33 Nowhere St.	111223333	D7	Dept. head	K2	Sales	10-1-2010	*
34Y70	Cheryl H. Clark	563 Downtown Ave.	999009999	F5	Floor manager	FM3	Sales	10-1-2009	*
23Y34	G. Jerry Smith	1555 Circle Dr.	111005555	S25X	Secretary	T5	Personnel	3-1-1999	4-30-2010
23Y34	G. Jerry Smith	1555 Circle Dr.	111005555	S26Z	Secretary	T6	Accounting	5-1-2010	*
.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.

Figure 9.5 An employee database consisting of three relations

EMPLOYEE relation

Empl Id	Name	Address	SSN
25X15	Joe E. Baker	33 Nowhere St.	111223333
34Y70	Cheryl H. Clark	563 Downtown Ave.	999009999
23Y34	G. Jerry Smith	1555 Circle Dr.	111005555

JOB relation

Job Id	Job Title	Skill Code	Dept
S25X	Secretary	T5	Personnel
S26Z	Secretary	T6	Accounting
F5	Floor manager	FM3	Sales
•	•	•	•
•	•	•	•
•	•	•	•

ASSIGNMENT relation

Empl Id	Job Id	Start Date	Term Date
23Y34	S25X	3-1-1999	4-30-2010
34Y70	F5	10-1-2009	*
23Y34	S26Z	5-1-2010	*
•	•	•	•
•	•	•	•
•	•	•	•

Figure 9.6 Finding the departments in which employee 23Y34 has worked

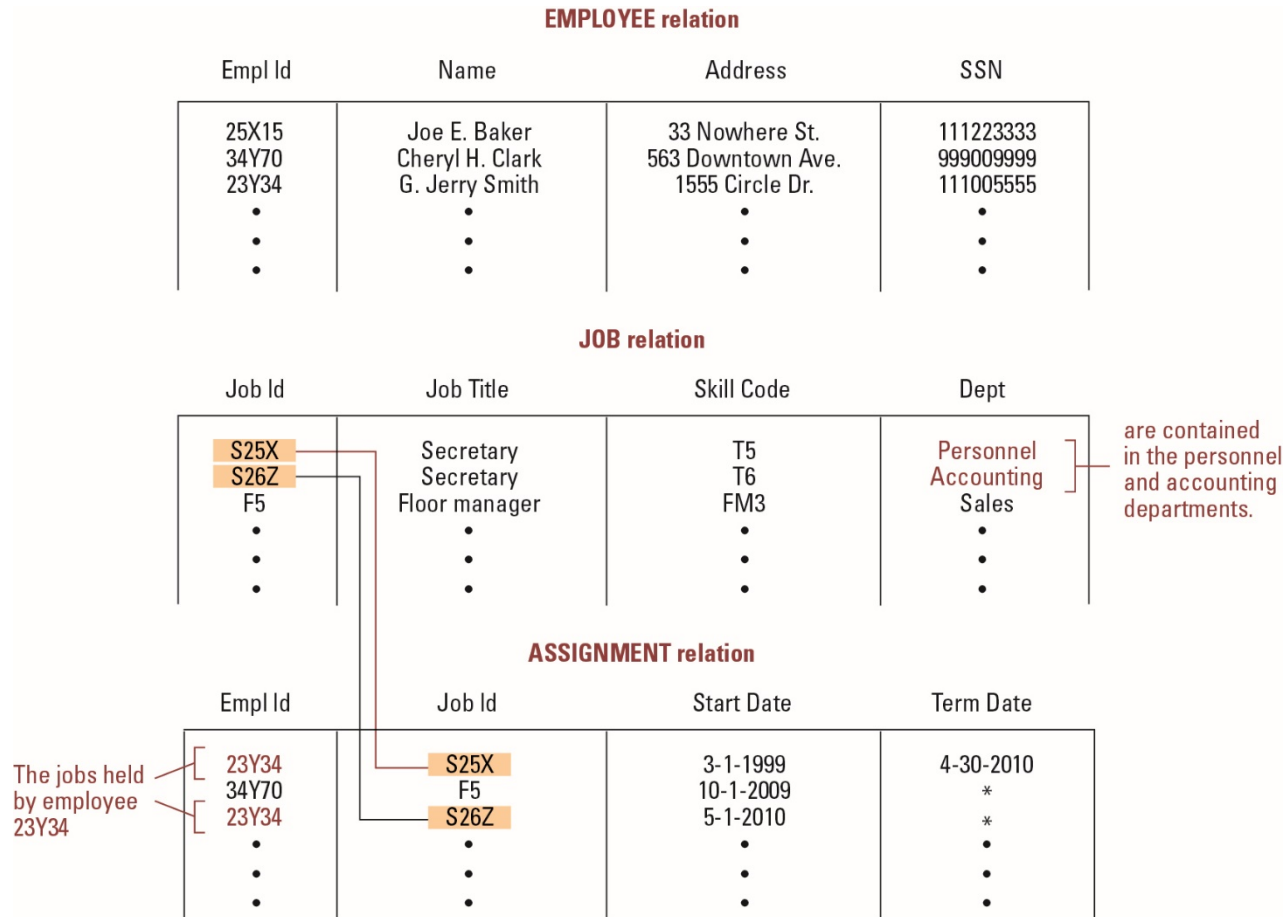
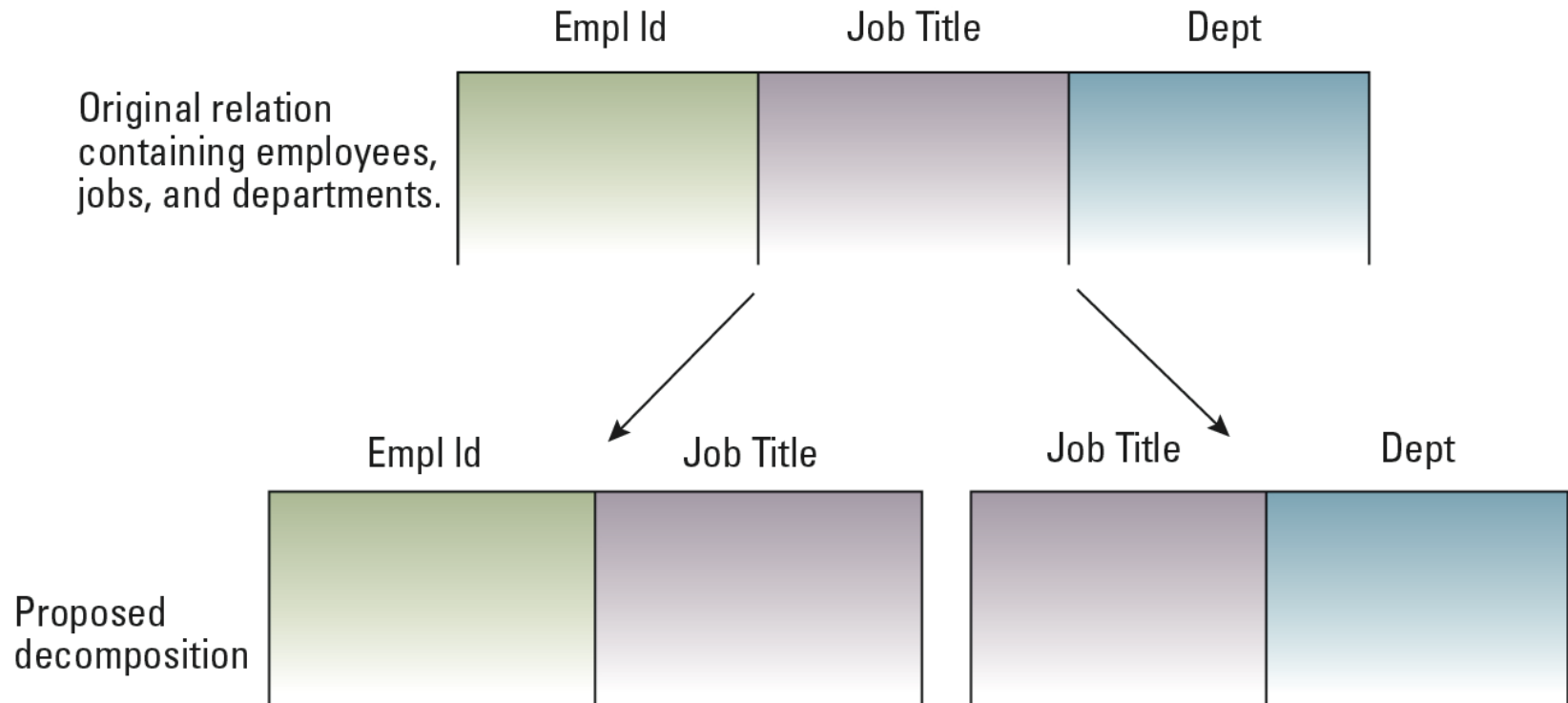


Figure 9.7 A relation and a proposed decomposition



Relational Operations

- **SELECT:** Choose rows
- **PROJECT:** Choose columns
- **JOIN:** Assemble information from two or more relations
組合來自兩個或多個關聯的資訊

Figure 9.8 The SELECT operation

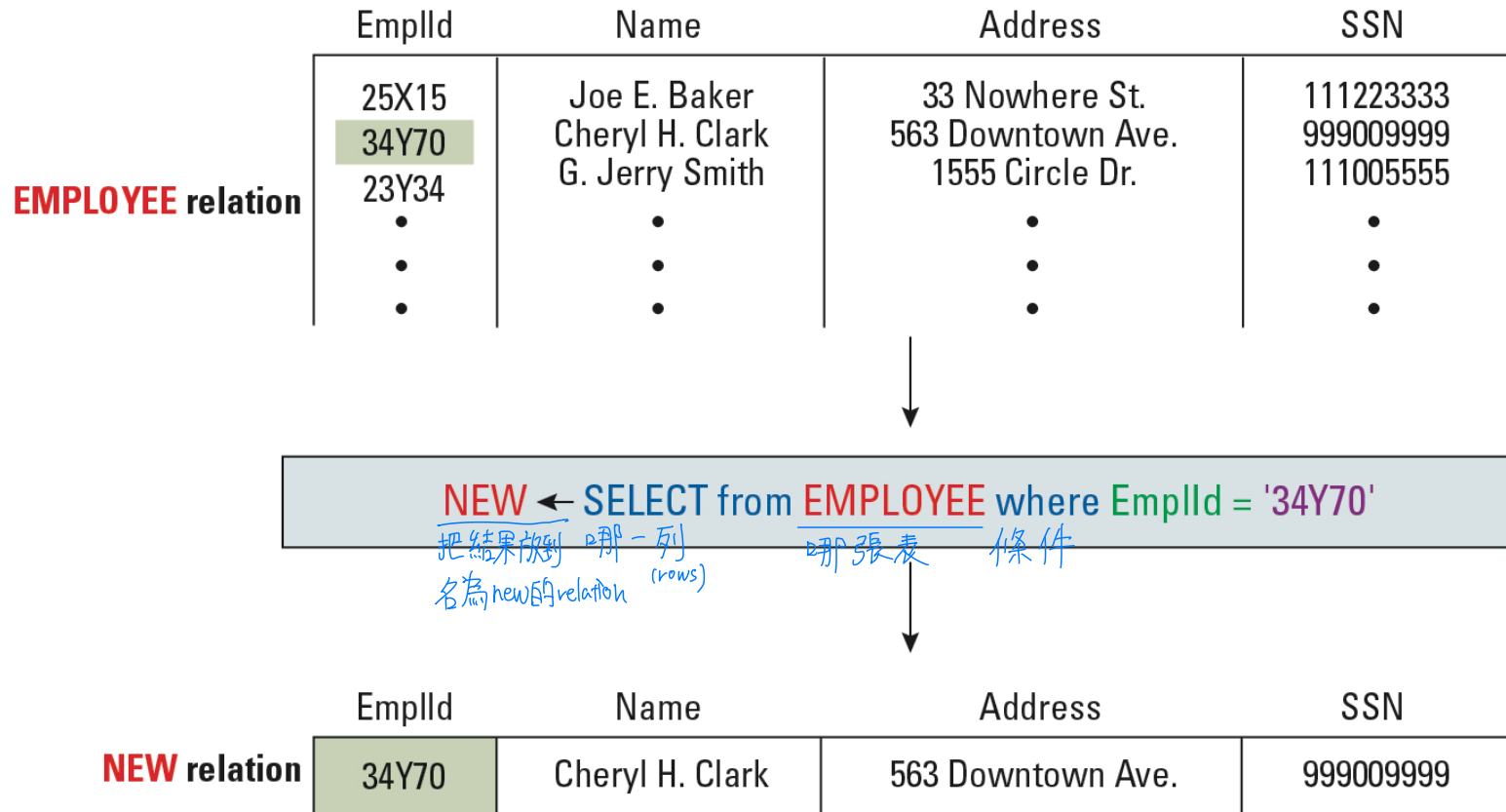


Figure 9.9 The PROJECT operation

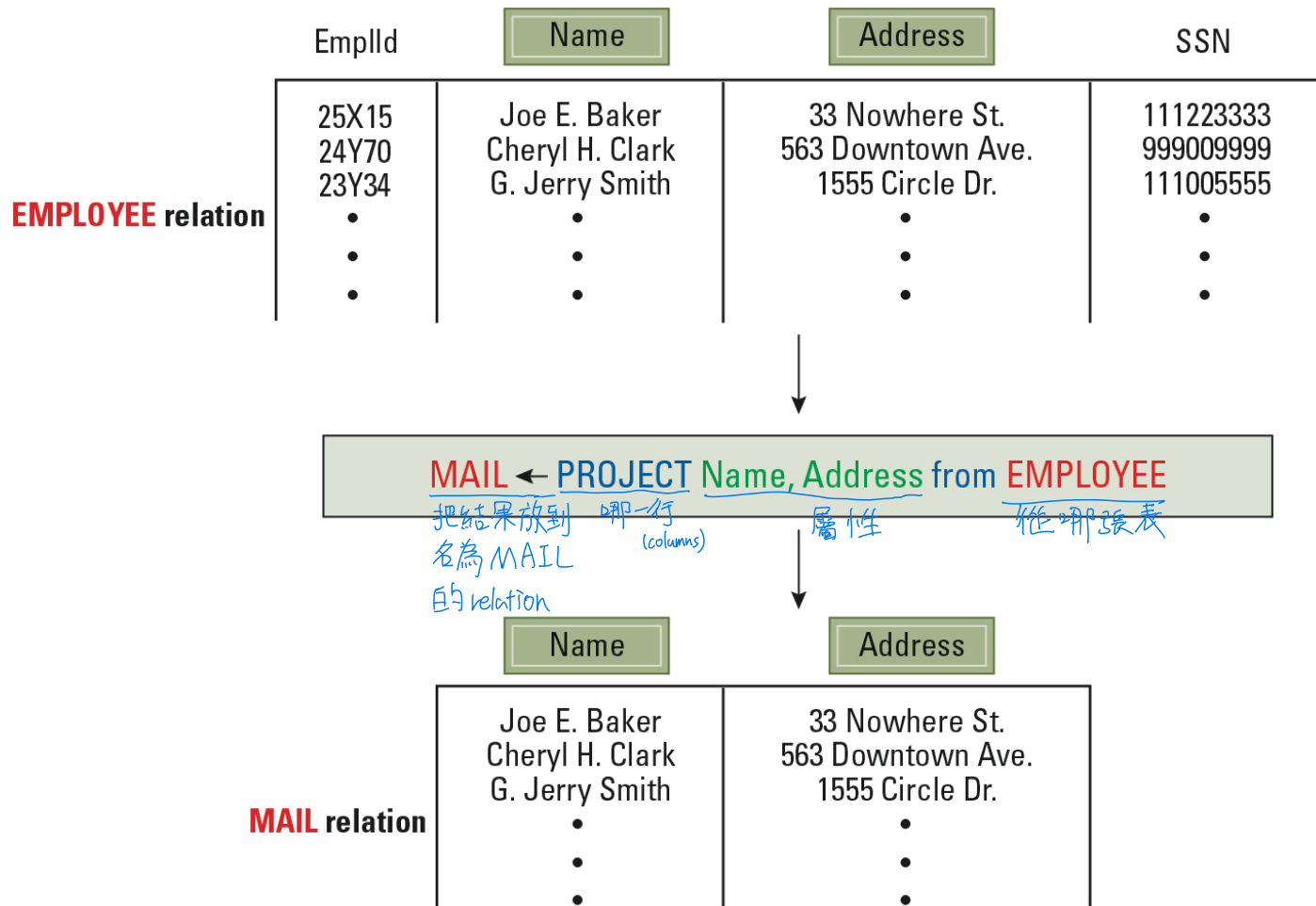


Figure 9.10 The JOIN operation 會考(要畫)

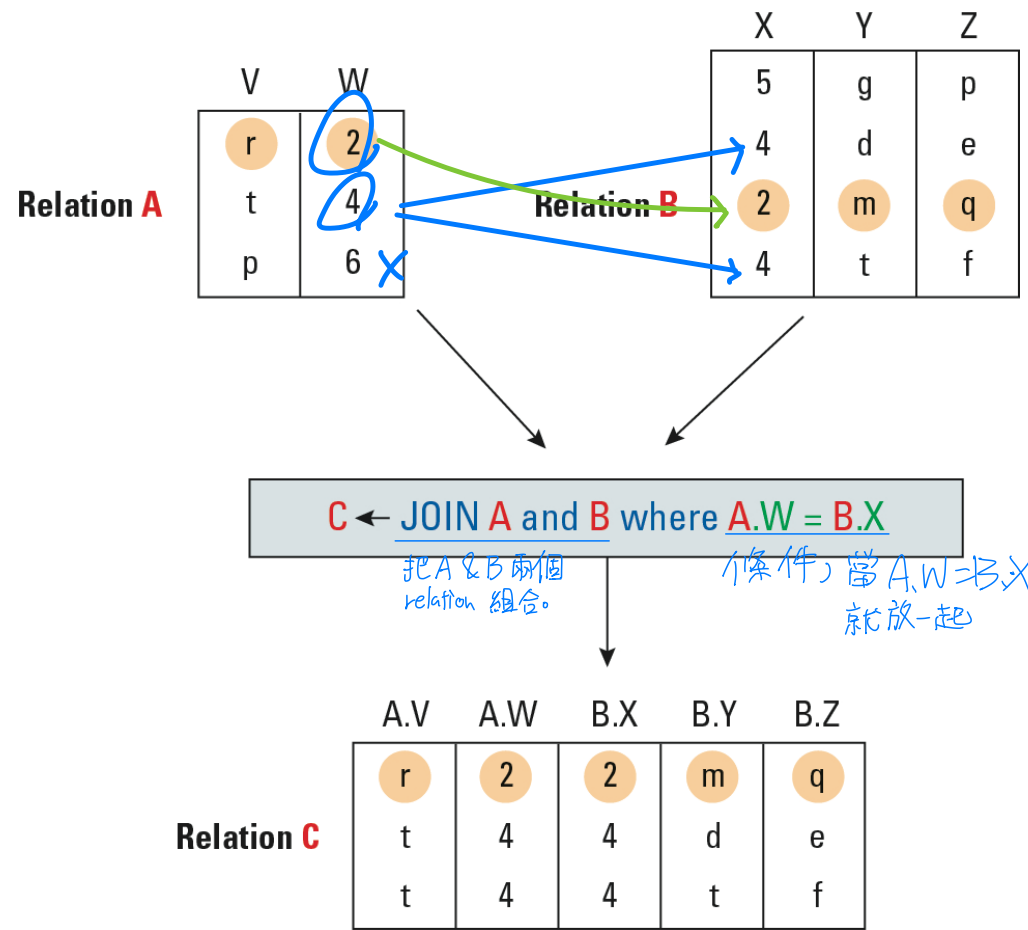


Figure 9.11 Another example of the JOIN operation

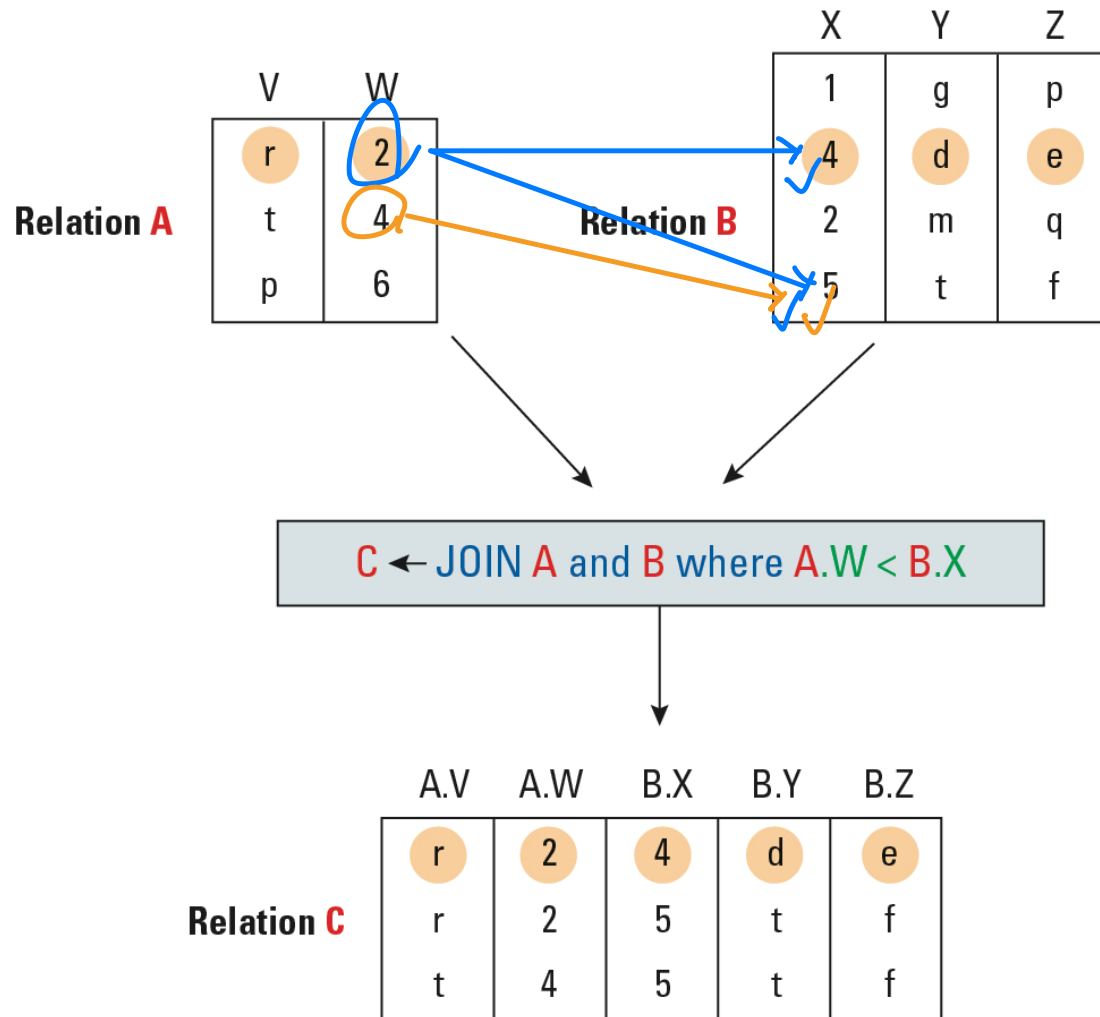
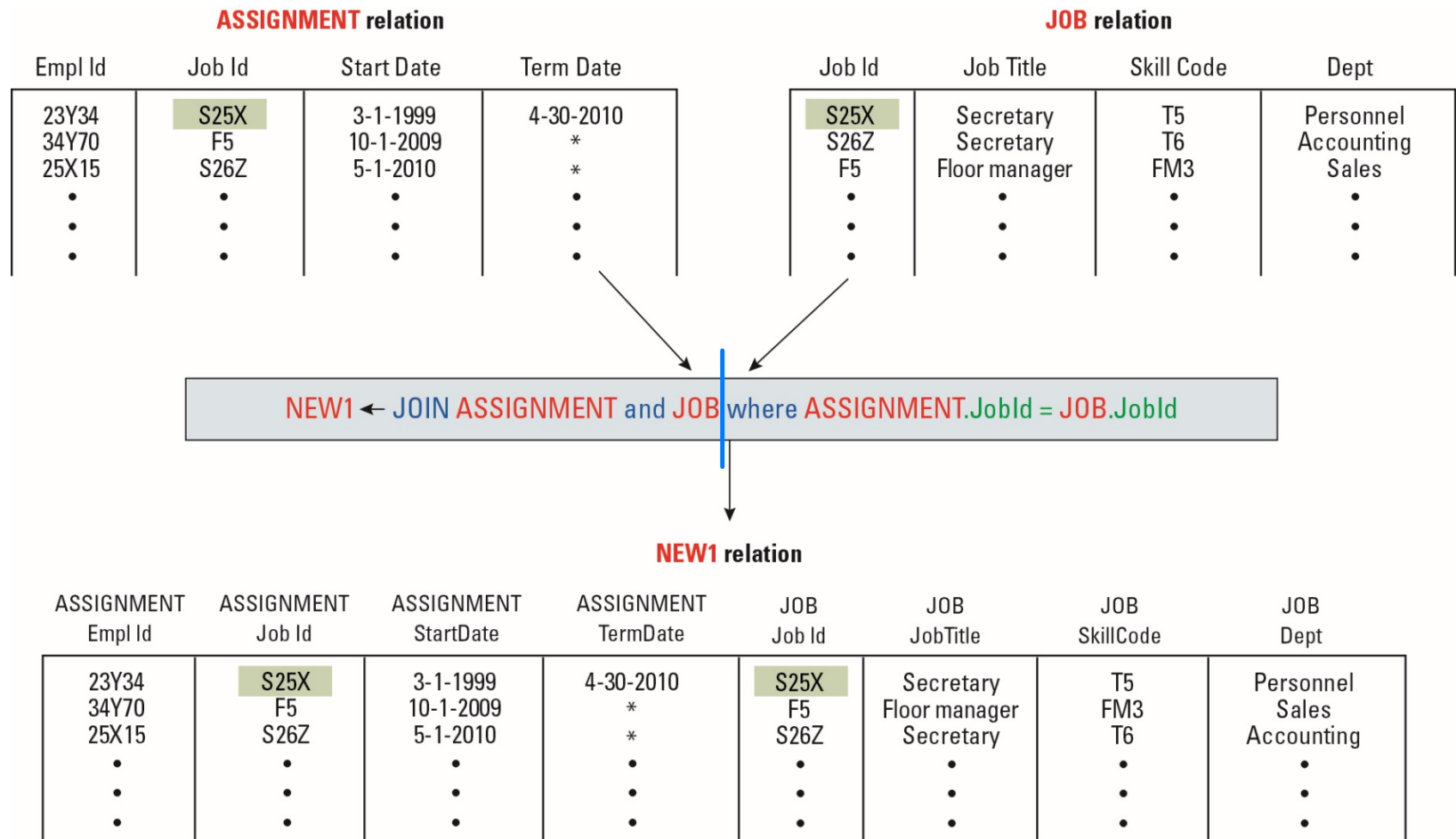


Figure 9.12 An application of the JOIN operation



Structured Query Language (SQL)

- Operations to manipulate tuples

- insert

常用SQL语法

- update

- delete

- select

SQL Examples

選 Rows

- SELECT EmplId, Dept
FROM Assignment, Job
WHERE Assignment.JobId = Job.JobId
AND Assignment.TermData = '*';

插入

Relation

- INSERT INTO Employee
VALUES ('43212', 'Sue A. Burt',
'33 Fair St.', '444661111');

Relation 中有幾個欄位、
有幾個值！

SQL Examples (continued)

刪除

- DELETE FROM Employee
WHERE Name = 'G. Jerry Smith';

更新

- UPDATE Employee
SET Address = '1812 Napoleon Ave.'
WHERE Name = 'Joe E. Baker';

9.3 Object-oriented Databases

物件导向資料庫

- **Object-oriented Database:** A database constructed by applying the **object-oriented paradigm**

- Each entity stored as a persistent object

每个实体儲存為永久性对象

- Relationships indicated by links between objects

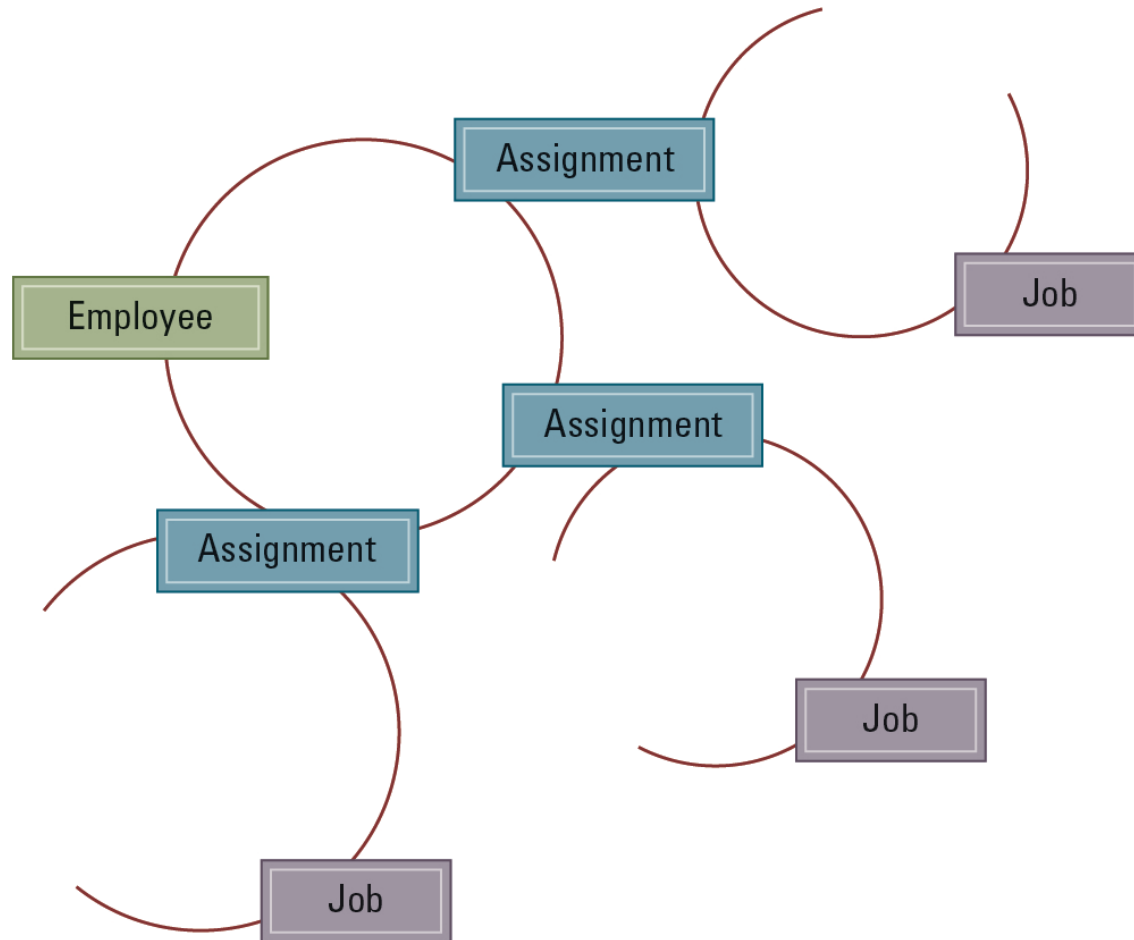
- DBMS maintains inter-object links

由DBMS来維護物件之間的連結

物件之間的連結
所表示的關係

→ 应用物件导向典範建立的資料庫

Figure 9.13 The associations between objects in an object-oriented database



Advantages of Object-oriented Databases

- Matches design paradigm of object-oriented applications 与 OOP 程式的设计準則相同!
- Intelligence can be built into attribute handlers
- Can handle exotic data types 可處理特殊資料類型
- Example: multimedia 多媒体

OOPDB 的優勢!

Intelligence: 自动化邏輯規則-算法

能在屬性處理器中加入一些特殊功能, 增加系統能力!

9.4 Maintaining Database Integrity 資料庫完整性

交互

- **Transaction:** A sequence of operations that must all happen together → 必須同時發生的操作

- Example: transferring money between bank accounts

交互日誌

- **Transaction log:** A non-volatile record of each transaction's activities, built before the transaction is allowed to execute 再執行任何交互前建構一個非揮發性的記錄。

- **Commit point:** The point at which a transaction has been recorded in the log 寫在交互日誌中該交互發生的時間點

- 回滾 **Roll-back:** The process of undoing a transaction

Maintaining database integrity (continued)

- Simultaneous access problems 同時訪問帶來的問題
 - Incorrect summary problem → 錯誤加總
 - Lost update problem → 遺失更新
- 鎖定 **Locking**: preventing others from accessing data being used by a transaction 防止其他 Statement 存取 Transaction 正在使用的資料
 - Shared lock: used when reading data 讀取時
 - Exclusive lock: used when altering data 修改時
(防止其他人同時寫入)

9.5 Traditional File Structures

必須按照順序來讀

- **Sequential file:** A file whose contents can only be read in order

- Reader must be able to detect end-of-file (EOF)

- Data can be stored in logical records, sorted by a key field 資料可存於邏輯記錄中，按關鍵字排序

- Greatly increases the speed of batch updates
提升批量更新的速度！

Figure 9.14 The structure of a simple employee file implemented as a text file

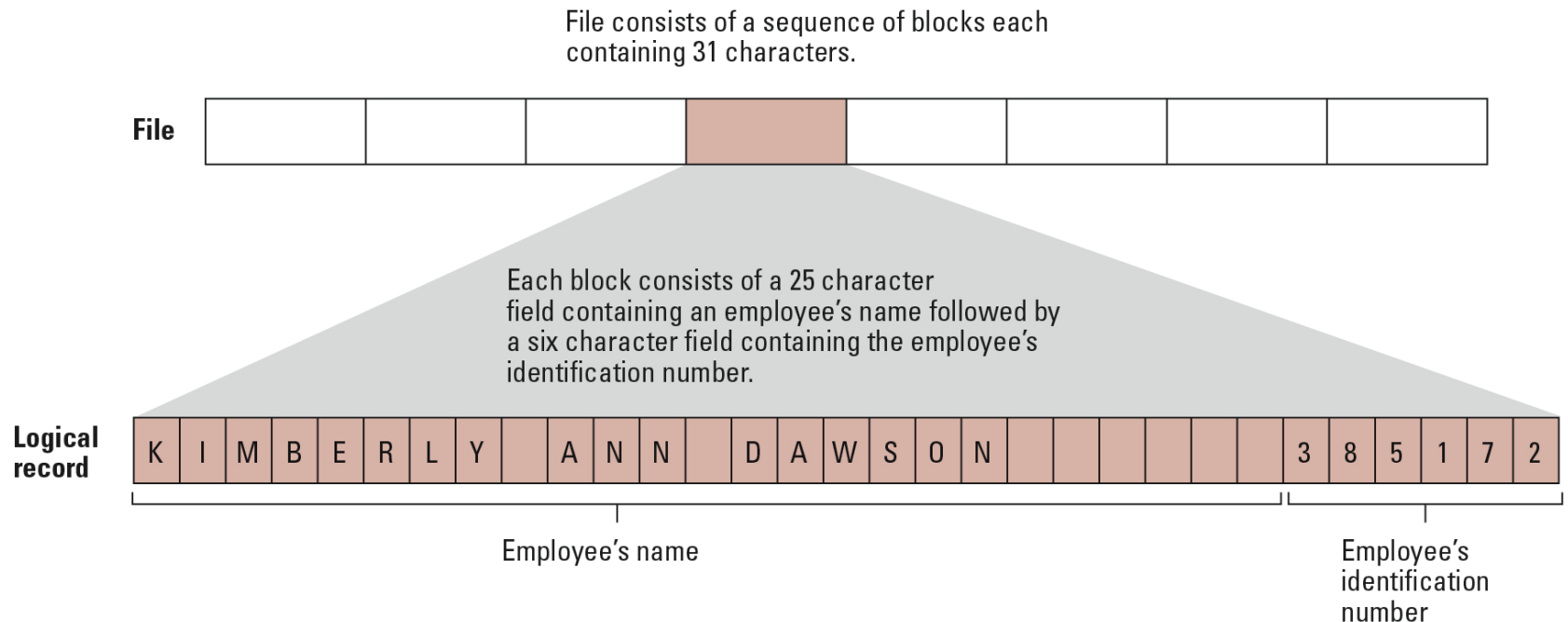
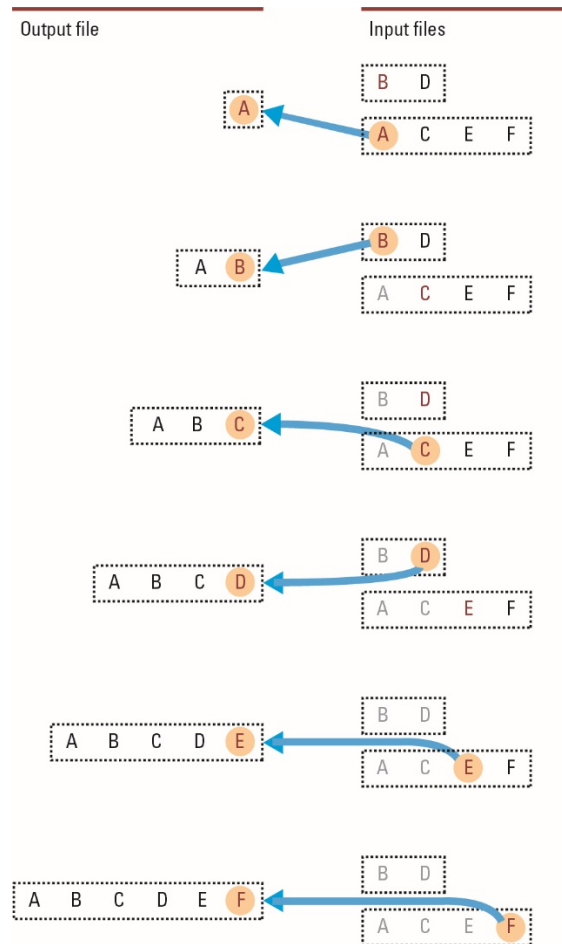


Figure 9.15 A function for merging two sequential files

课P.513

```
def MergeFiles (InputFileA, InputFileB, OutputFile):  
    if (both input files at EOF):  
        Stop, with OutputFile empty  
  
    if (InputFileA not at EOF):  
        Declare its first record to be its current record  
  
    if (InputFileB not at EOF):  
        Declare its first record to be its current record  
  
    while (neither input file at EOF):  
        Put the current record with the “smaller” key field value in OutputFile  
        if (that current record is the last record in its corresponding input file):  
            Declare that input file to be at EOF  
        else:  
            Declare the next record in that input file to be the file’s current record  
  
    Starting with the current record in the input file that is not at EOF,  
    copy the remaining records to OutputFile
```

Figure 9.16 Applying the merge algorithm (Letters are used to represent entire records. The particular letter indicates the value of the record's key field.)

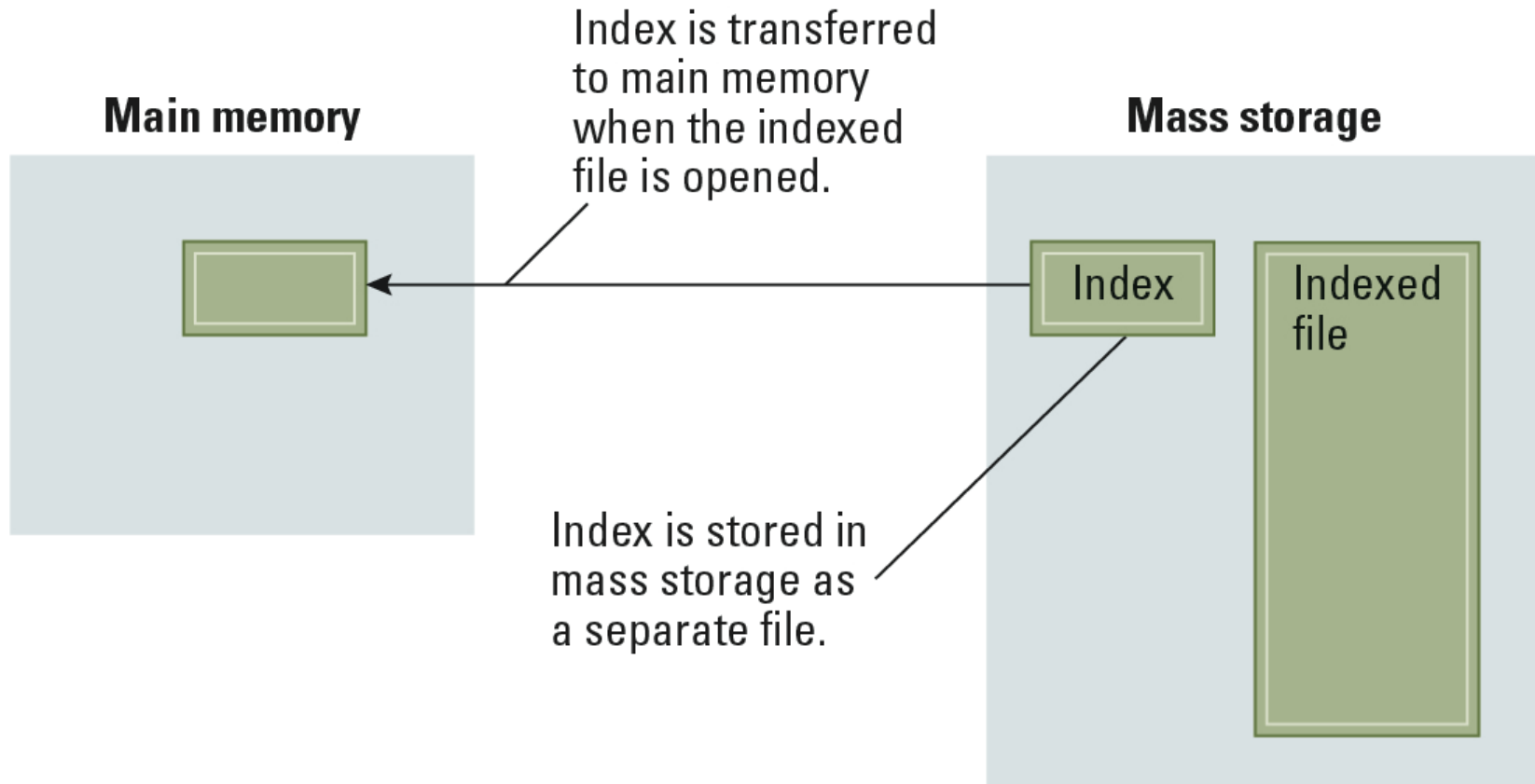


Indexed Files 索引文件

快速搜尋

- **Index:** A list of key values and the location of their associated records → 包含鍵值 & 相關位置記錄的列表
- Efficient way to quickly identify the location of a desired record 快速尋找所需記錄位置的方法
- Index is stored as a separate file
單獨的文件

Figure 9.17 Opening an indexed file



Hash Files 哈希资料(雜湊)

- Each record has a key field 每筆記錄都有一鍵值
- The storage space is divided into **buckets** 資料被存在"儲存桶"
- A **hash function** computes a bucket number for each key value 由hash func 計算出的鍵值來決定要放在哪個儲存桶
- Each record is stored in the bucket corresponding to the hash of its key 每筆資料都存在與其鍵值之Hash值对应的儲存桶中。

Figure 9.18 Hashing the key field value 25X3Z to one of 41 buckets

\ Assume Hash function is $x \bmod 41$

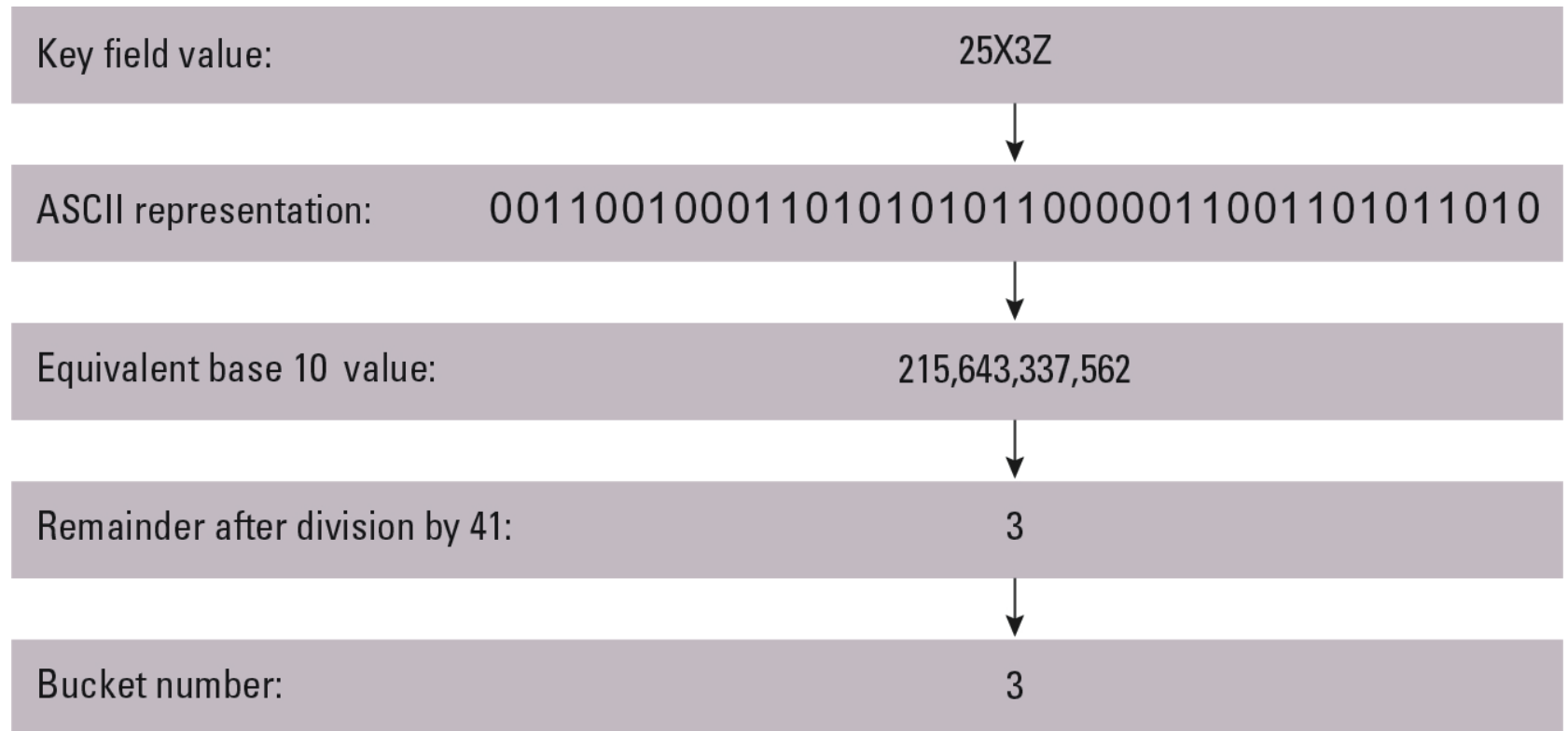
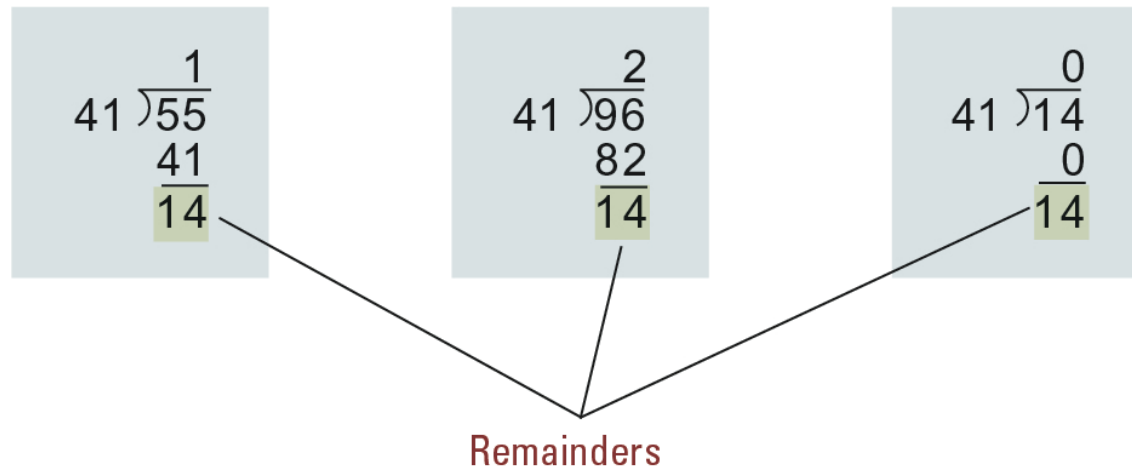
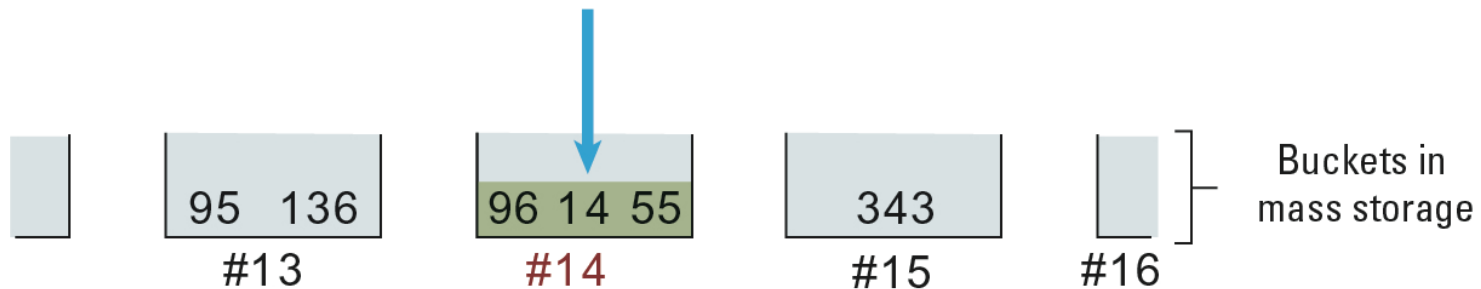


Figure 9.19 The rudiments of a hashing system



When divided by 41, the key field values of 14, 55, and 96 each produce a remainder of 14. Thus these records are stored in bucket 14.



Collisions in Hashing

碰撞

- **Collision:** The case of two keys hashing to the same bucket → 兩個鍵值經Hash後對應到相同的桶

- Major problem when table is over 75% full

- Solution: increase number of buckets and rehash all data

當表的剩餘空間剩下25%時會出現的問題

期末考 考到這！

Range: 4.2 ~ 9.5

Chapter. 7 跳過, 會考SQL語法

9.6 Data Mining

- **Data Mining:** The area of computer science that deals with discovering patterns in collections of data
- **Data warehouse:** A static data collection to be mined
- **Data cube:** Data presented from many perspectives to enable mining

Data Mining Strategies

- Class description
- Class discrimination
- Cluster analysis
- Association analysis
- Outlier analysis
- Sequential pattern analysis

9.7 Social Impact of Database Technology

- Problems
 - Massive amounts of personal data are being collected
 - Often without knowledge or meaningful consent of affected people
 - Data merging produces new, more invasive information
 - Errors are widely disseminated and hard to correct
- Remedies
 - Existing legal remedies often difficult to apply
 - Negative publicity may be more effective